RIG-SMOOTH HOMOMORPHISMS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-base-change-rig-smooth-homomorphism; the proof environment immediately following it is its own proof

Editorial note. The label lemma-base-change-rig-smooth-homomorphism names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the base change setup for pairs (A1, I1) to (A2, I2) of Noetherian rings with V(I1 A2) = V(I2) and the base change algebra B2 (remark-base-change, equation-C-prime)
- the Ext^1 annihilation criterion and its persistence under base change (definition-rig-smooth-homomorphism and lemma-zero-ext-1-after-modding-out, delivered as its own card)
- commensurability of ideals with equal radical in a Noetherian ring: I2^c is contained in I1 A2 for some c >= 0

Source reference. Original reference 0GAI: sources/stacks-restricted/pinned-original-source.tex, source lines 831–831. Prerequisite pointer invoked at 1055; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at 1046; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAL: sources/stacks-restricted/pinned-original-source.tex, source lines 1016–1016. Prerequisite pointer invoked at 1054; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06FQ: sources/stacks-restricted/pinned-original-source.tex, source lines 426–426. Prerequisite pointer invoked at 1048; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $A_1 \to A_2$ be a map of Noetherian rings. Let $I_i \subset A_i$ be an ideal such that $V(I_1A_2) = V(I_2)$. Let B_1 be in (0AL4) for (A_1, I_1). Let B_2 be the base change of B_1 as in Remark 06FQ. If B_1 is rig-smooth over (A_1, I_1), then B_2 is rig-smooth over (A_2, I_2).*

Proof. Follows from Lemma 0GAL and Definition 0GAI and the fact that I_2^c is contained in I_1A_2 for some $c \geq 0$ as A_2 is Noetherian. □